NSE Stock Analyzer

User guide — installation, daily use, and how to read every screen

Technical analysis for Indian stocks and indices across three timeframes, an ML model trained on each stock's own history, a candlestick-pattern model, automatic daily data from NSE bhavcopy, and a ChromaDB memory of every analysis.

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For educational purposes only. This app is not investment advice or a recommendation, and is not a SEBI-registered investment adviser or research analyst. Its analyses and predictions are generated automatically and can be wrong. 9 out of 10 individual traders in equity F&O incurred net losses (SEBI study, January 2023). Always do your own research, manage your risk, and consult a SEBI-registered adviser before investing.

Contents

1. What the app does
2. Installing — Docker (recommended)
3. Installing — without Docker
4. First run and the guided tour
5. The morning briefing (main page)
6. Getting data in: CSV upload
7. Getting data in: Watchlist & auto-sync
8. Reading an analysis
9. History & semantic search
10. Your data, updates and backups
11. Using the app with Claude (skill file)
12. Troubleshooting
13. Settings reference

1. What the app does

Give it price history for an NSE stock or index (by uploading a CSV, or by letting it download NSE's daily bhavcopy files) and it produces a full analysis:

- **A morning briefing as the main page** — built automatically every weekday at 08:15 IST (before 08:30): world indices, commodities and currencies; Indian indices, sectors, market breadth and a summary of every NSE index; India and world news summaries; today's and this week's stock events; and a pre-market outlook for the Nifty. A weekly recap is built every Saturday.
- **Three timeframe views** — Day trading (next session), Short term (1–4 weeks) and Long term (months), each with a verdict, a score from −100 to +100, the indicator signals behind it and a trade plan (entry, stop-loss, targets).
- **An ML model per stock** — a gradient-boosted classifier trained on that stock's own history predicts the chance of a rise over 1, 5 and 20 sessions. Its accuracy is tested walk-forward against a simple baseline, and it only gets real weight in the verdict when it beats that baseline.
- **Candlestick patterns** — 22 patterns detected on every candle, with how each has performed on this stock and an estimated success chance for today's pattern.
- **Charts** — candlesticks with overlays, support/resistance and pattern markers; RSI, MACD, Stochastic and ADX panels.
- **Memory** — every analysis is stored in ChromaDB. The next analysis compares with the last one, checks whether earlier calls came true, and finds similar setups in other stocks.
- **Automatic daily updates** — a watchlist that downloads each day's NSE bhavcopy at 18:45 IST and re-analyses your stocks.

Disclaimers in the app

Every time the app is opened, it shows a notice that it is **for educational purposes only**, together with SEBI's risk disclosure for derivatives traders (from SEBI's study dated January 25, 2023: 9 out of 10 individual traders in equity F&O incurred net losses). You must accept it before using the app. A short version stays in the header and footer of every screen; **Read disclaimer** opens the full text again.

Realistic expectations

Day-to-day stock direction is close to a coin flip. A model that is right 52–56% of the time in honest testing is doing well, and on many stocks it will show no edge at all. The app tells you this openly instead of hiding it — look at the accuracy-versus-baseline line under every model output and at the track record in the Memory panel.

2. Installing — Docker (recommended)

Docker runs the backend and frontend in containers, so you don't need to install Python or Node yourself.

Requirements

- Docker Engine with the Compose plugin (Linux), or Docker Desktop (Windows / macOS).

Start the app

1. Unzip the project and open a terminal in its root folder (the one containing **docker-compose.yml**).
2. Create the data folder first so it is owned by you (Linux/macOS):

```
mkdir -p ~/nse-analyzer-data
```

3. Build and start everything:

```
docker compose up -d --build
```

4. Open **http://localhost:8080** in your browser. The API and its interactive docs are at **http://localhost:8000/docs**.

Everyday commands

Task	Command
Stop	docker compose down
Start again	docker compose up -d
Update to a new version (after replacing the code)	docker compose up -d --build
See backend logs (sync progress, errors)	docker compose logs -f backend
Run a one-off sync	docker compose exec backend python scripts/daily_sync.py

Docker settings (optional)

Copy **.env.example** to **.env** next to docker-compose.yml to change defaults:

Setting	Default	Meaning
NSE_DATA_HOST_DIR	~/nse-analyzer-data	Folder on your computer where all data is kept
WEB_PORT	8080	Port for the web UI
AUTO_SYNC_TIME	18:45	Daily sync time (IST, Mon–Fri)
BRIEFING_TIME / WEEKLY_TIME	08:15 / 09:00	Morning briefing (Mon–Fri) and weekly recap (Sat), IST
HOST_UID / HOST_GID	1000 / 1000	Your user/group id (run **id -u** and **id -g**) so data files stay owned by you

The containers restart automatically after a reboot (**restart: unless-stopped**), so the daily 18:45 sync keeps running as long as Docker is running.

3. Installing — without Docker

Requirements: **Python 3.11 or 3.12** and **Node.js 18+**. You need two terminals: one for the backend, one for the frontend.

Backend — Linux / macOS

```
cd nse-analyzer/backend
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
```

Backend — Windows (PowerShell)

```
cd nse-analyzer\backend
python -m venv .venv
.venv\Scripts\activate
python -m pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
```

If PowerShell says running scripts is disabled, run this once and activate again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Frontend (any OS)

```
cd nse-analyzer/frontend
npm install
npm run dev
```

Then open **http://localhost:5173**.

Virtual-environment tips

- You'll see **(.venv)** at the start of the prompt when it is active. In each new terminal, only the activate line is needed.
- **deactivate** leaves the environment.
- If you deleted an old .venv and get “cannot execute: required file not found”, open a fresh terminal (or run **deactivate** then **hash -r**), delete .venv and create it again.
- On Ubuntu/Debian, “ensurepip is not available” means you need **sudo apt install python3-venv**.

4. First run and the guided tour

The first time you open the app, a guided tour starts automatically. It walks through every screen in about two minutes, spotlighting each part — starting with the morning briefing — and explaining what it shows. Use **Next** / **Back** or the arrow keys; **Esc** or **Skip tour** closes it.

- **No data yet?** At step 4 the tour offers **Load sample & continue**, which analyses a bundled demo stock so you can see every panel. Sample analyses are never saved to your history or memory.
- **Replay any time** with the **Tour** button at the top right.
- The header also shows how many analyses are stored and the exact folder your data lives in.

5. The morning briefing (main page)

The **Market** tab opens first. Every weekday at **08:15 IST** — before the 08:30 pre-market and the 09:15 open — the app downloads the overnight data, summarises it on your own computer and saves a briefing. If the computer was off at 08:15, the briefing is built as soon as the backend starts. **Refresh now** rebuilds it at any time; the date picker opens earlier briefings.

Where the data comes from

Data	Source
World indices, crude, gold, silver, USD/INR, dollar index, US 10-year yield	Yahoo Finance
World news	CNBC (top news, world, Asia, economy) and Yahoo Finance
Indian market news and economy	CNBC-TV18 (markets, economy, world)
Indian indices, all-index summary	NSE live snapshot; NSE's official daily index file as fallback
Market breadth and top movers	NSE bhavcopy (the same files the Watchlist uses)
Results, dividends, bonus, splits, board meetings	NSE event calendar and corporate actions
FII / DII buying and selling (cash market)	NSE FII/DII provisional data
F&O; positions of Retail, Pro, FII and DII	NSE participant-wise open interest file (public, daily)

What each part of the daily briefing shows

Section	What it tells you
Global cues	Average overnight move of US and Asian indices, labelled positive, mildly positive, flat or negative.
News mood	Sentiment of the India and world stories, scored offline, from −100 to +100.
Pre-market outlook	The model's chance that the Nifty 50 opens higher and closes higher than yesterday. **Why?** lists the overnight moves that pushed it. The line beneath shows its tested accuracy against a naive guess — if it says “no reliable edge”, treat the number as indicative only. Probabilities are shown between 1% and 99%.
Summary	Everything below in a few plain-English paragraphs.
World markets	US, Europe and Asia indices plus commodities and currencies, each with a 1-month trend. Grey rows are older data (a market holiday, for example).
Indian markets	Nifty 50, Sensex, Nifty Bank, Midcap, Smallcap, India VIX; sector moves; breadth (advances vs declines across all NSE stocks); top gainers and losers among stocks with turnover above ₹25 crore.
All NSE indices	Every NSE index — broad market, sectoral, thematic, strategy and fixed income — with change, P/E, position in its 52-week range, advances/declines and 1-year change. Filter by group, search (for example “defence”) and sort.

Section	What it tells you
Who's buying & selling	FII and DII net buying or selling in the cash market (₹ crore, provisional) with a 10-session chart. Below it, the futures & options positions of Retail (NSE's “Client” category), Pro traders (brokers trading their own money), FIIs and DIIs: the share of each group's index-futures position that is long, net contracts and the change from the previous session, net index calls and puts, and net stock futures. A chart shows how long each group has been over the last 10 sessions. These are positions at the end of the last session, not a forecast.
Advance / decline	How many NSE stocks rose, fell or were unchanged, the A/D ratio, how many moved 4% or more, advances minus declines for each of the last 10 sessions, and advances vs declines inside each index.
News summary	Stories from the last 30 hours. The same story from several sources is shown once; stories are grouped into themes, each with a sentiment tag. ▲ positive, ▼ negative, ● neutral.
Stocks in the news	Companies named in headlines, ★ for your watchlist. Click one for its headlines.
Stock events	Today's and the next 7 days' board meetings (results, dividends, fund raising) and ex-dates. Watchlist stocks are starred and listed first.

Weekly recap

Built every Saturday at 09:00 IST (switch with **Weekly**). It shows the week's change for world and Indian indices, sectors and every NSE index; FII/DII net for the week and how each participant's F&O; positions changed; advance/decline by session; the news mood day by day and the recurring themes; how the pre-market calls compared with the actual opening; and the week ahead's results and corporate actions. It is built from the morning briefings stored that week, so it gets richer once the app has run for a full week.

How it works offline

- **Sentiment** — a finance word list with phrase and negation rules (“misses estimates”, “not higher”). Optional: set SENTIMENT_BACKEND=finbert after installing transformers and torch to use the FinBERT model locally.
- **Themes** — the same story from several feeds is merged by text similarity; stories are then clustered into themes named after their subject (a company, a country, a policy), not after words like “rises”.
- **Key stories** — TextRank picks the most central headlines.
- **Company linking** — headlines are matched to NSE symbols using NSE's company list.
- **Pre-market outlook** — a logistic regression trained on about five years of history: yesterday's moves in US, European and Asian indices, crude, gold, the dollar, USD/INR and US yields, tested walk-forward.

If NSE blocks a request (it sometimes does), the rest of the briefing still builds. The “sources” line at the bottom of the page lists what loaded. The all-index summary then falls back to NSE's official daily file.

6. Getting data in: CSV upload

Download a CSV from NSE

1. Go to nseindia.com, search the stock and open its quote page.
2. Open **Historical Data**, choose a range (1 year or more is recommended) and click **Download (.csv)**.
3. For indices (NIFTY 50, NIFTY BANK…) use the indices historical-data page; those files work too.

Upload it

1. Drop the file on the upload area (or click to browse). The symbol is detected from the file name; you can type it instead.
2. Leave **Merge with stored history** ticked — each upload is added to the stock's stored history, so the training data grows over time.
3. Click **Analyse**. It takes a few seconds while the models train.

Daily uploads

After the first year of history, you can upload just one new day each evening — a one-row file is fine. It is merged with what's stored, all models retrain on the full history, and the previous day's calls are checked against what actually happened. Overlapping or repeated days are handled automatically. If you'd rather not do this by hand, use the Watchlist (next section).

Also accepted: NSE bhavcopy files containing every stock (type the symbol you want) and generic OHLCV CSVs, daily or intraday.

7. Getting data in: Watchlist & auto-sync

The Watchlist downloads NSE's public end-of-day bhavcopy files for you, so you never have to upload anything.

Add stocks

1. Open the **Watchlist** tab.
2. Type NSE symbols separated by commas — e.g. **RELIANCE, TCS, M&M, BAJAJ-AUTO, NIFTY 50**. Use the exact NSE ticker; indices can be written with or without spaces.
3. Choose how much history to backfill for new symbols (6 months, 1 year or 2 years) and click **Add & sync**.
4. A progress bar and log show what's happening. The first 1-year backfill downloads about 250 daily files and takes a few minutes; files are cached, so adding more stocks later is almost instant.

Stocks you already uploaded by CSV simply continue from their last date — nothing is downloaded twice.

The buttons

Button	What it does
Sync now	Downloads any missing trading days for every symbol, then analyses the ones that got new data.
Analyse all	Re-runs the analysis for every symbol using data already stored — no download.
Analyse (row)	Fresh analysis of that one symbol from stored data; opens the result.
Open (row)	Shows the last saved analysis without re-running it.
↻ (row)	Syncs just that symbol.
× (row)	Stops tracking the symbol. Its stored history and past analyses are kept.

Automatic daily sync

- Runs Monday–Friday at **18:45 IST** (NSE publishes the bhavcopy around 6 pm) while the backend is running.
- If the computer was off, the next start catches up on the missed days automatically.
- Holidays are detected automatically (no file that day).
- If NSE serves a file with the wrong date, it is ignored and retried at the next sync — the log says so.
- Prefer not to keep the server running? Schedule the one-shot script with cron or Windows Task Scheduler:

```
cd nse-analyzer/backend && .venv/bin/python scripts/daily_sync.py

# cron entry, weekdays at 18:45:
45 18 * * 1-5  cd /path/to/backend && .venv/bin/python scripts/daily_sync.py
```

8. Reading an analysis

Every analysis — from an upload, a sync or the Analyse button — opens on the Analyse screen, top to bottom in the order below.

Header and summary

- **Header** — symbol, latest close and day change, where the data came from (CSV upload, NSE auto-sync, stored data or sample), how many sessions the models were trained on, and when the analysis ran.
- **Summary** — everything on the screen condensed into one plain-English paragraph. This is also the text saved to ChromaDB memory and used by semantic search.

Timeframe cards

Element	Meaning
Verdict badge	Strong Bullish (≥ +60), Bullish (≥ +25), Neutral, Bearish (≤ −25), Strong Bearish (≤ −60), from the combined score.
Score bar	Where the combined score sits between −100 (left) and +100 (right).
Combined	Final score: the technical score blended with the ML model.
Technical	Rule-based score alone — the average vote of that timeframe's indicators × 100.
ML P(up)	Model's probability the price is higher after 1 / 5 / 20 sessions.
Model line	Walk-forward accuracy versus baseline. **“Has an edge”** means it beat the baseline by more than 2 points and gets 40% weight; otherwise 15%.
Trade plan	Bias, entry, stop-loss, targets and risk:reward from ATR, pivots and support/resistance. A starting point for your judgement, not advice.

Example: the day-trading model predicts an 86% chance of a rise, but the technical rules score −22. Because the model only has a modest edge, the combined score lands at +15 — Neutral. When the two disagree like this, treat the view with caution.

Why this verdict?

Click **Show N signals** under any card to see every indicator that voted for that timeframe:

- ▲ bullish vote, ▼ bearish vote, ● neutral.
- Each line shows the indicator, its current reading and a one-line reason — e.g. “Closed below pivot”, “EMA9 below EMA21”, “Lower-value CPR vs previous session”.
- The technical score is simply the average of these votes × 100, so you can always see exactly what drove the verdict.

Day trading uses pivots/CPR, EMA 9/21, RSI, Stochastic, Supertrend, VWAP and candle patterns. Short term uses EMA 21/50, MACD, RSI, Bollinger Bands, ADX and volume. Long term uses the 50/200-day averages, 52-week range, 6/12-month returns and OBV.

Price chart

- Toggle overlays with the chips: SMA 20/50/200, EMA 9/21, VWAP, Bollinger Bands, Supertrend, S/R levels and candle patterns.
- Green arrows below candles mark bullish patterns; red arrows above mark bearish ones. Hover a candle to see its OHLC values and pattern names.

- Scroll to zoom, drag to pan. R1–R3 / S1–S3 lines are the nearest swing resistance and support levels.

Candlestick patterns

- **Pattern says / Success chance / Reliability** — the direction today's pattern points, and the estimated chance it works within 3 sessions. It blends this stock's own record with a model trained on pattern events from every stock you track. Low reliability means don't trade the pattern on its own.
- **How patterns performed on this stock** — how often each pattern was followed by a move in its direction after 3 sessions, and the average move after 3 and 10 sessions.
- **Recent signals** — the latest patterns with the actual move afterwards (✓ worked, ✗ failed, or pending).

Indicators

Panel	How to read it
RSI (14)	Above 70 overbought, below 30 oversold; above 50 means buyers are in control.
MACD	MACD above its signal line is bullish momentum; a growing histogram means momentum is strengthening.
Stochastic	Above 80 overbought, below 20 oversold; %K crossing %D signals a turn.
ADX / DMI	ADX above 25 means a real trend; +DI above −DI means the trend is up.

Use 3M / 6M / 1Y / All to change the range.

Levels and model probabilities

- **Pivots & CPR** — classic pivot levels for the next session. A narrow CPR (central pivot range) often precedes a trending day; a wide one a sideways day.
- **Support & resistance** — swing levels from the last year with how many times each was tested.
- **Model probability of rise** — the ML probability for each timeframe; the dashed line marks 50%.

Memory

Track record is the most honest measure of usefulness: each earlier Bullish or Bearish call is checked once enough new sessions arrive (1, 5 or 20). Neutral calls are skipped.

9. History & semantic search

Pick a symbol to see all its stored analyses, newest first, with a chart of how the three timeframe scores have moved. **Delete** removes that symbol's stored analyses from memory (price history is kept).

Search in plain words — for example oversold bounce near support or bearish engulfing near resistance — and the app finds the most similar past analyses across all your stocks, using ChromaDB embeddings.

10. Your data, updates and backups

Everything the app builds up is stored **outside the project folder**, in **~/nse-analyzer-data** (Linux: /home/<you>/nse-analyzer-data · Windows: C:\Users\<you>\nse-analyzer-data). Docker uses the same folder.

Folder / file	Contents
history/	Merged price history per symbol — the models' training data
watchlist.json	Your watchlist
results/	Latest full analysis per symbol (what Open shows)
chroma_db/	ChromaDB memory of every analysis
models/	Trained ML and pattern models
bhavcopy/	Cached NSE daily files (shared by all symbols)

- **Updating the app:** replace the project folder with the new zip. Your data is untouched. With Docker, run **docker compose up -d --build**; without Docker, re-run **pip install -r requirements.txt** and restart.
- **Backup:** copy the nse-analyzer-data folder.
- **Different location:** set NSE_DATA_DIR (or NSE_DATA_HOST_DIR for Docker).
- **Upgrading from an older version** that kept data in backend/data: the first start copies it to the new folder automatically and leaves a MOVED_TO.txt note.

11. Using the app with Claude (skill file)

The project includes **skills/nse-stock-analyzer/SKILL.md**, which teaches Claude how to operate the app through its API: add symbols, run syncs, analyse, read results honestly (including whether the models have an edge) and troubleshoot. Add the folder as a skill in Claude, or place it in your project's skills directory for Claude Code; then ask things like “How does the market look this morning?” or “Add HDFCBANK to my watchlist and tell me the short-term view.” The app must be running for Claude to reach it.

12. Troubleshooting

Problem	Fix
Header says “backend not reachable”	Start the backend (uvicorn, or docker compose up -d). Check it at http://localhost:8000/api/health. With Docker run docker compose logs backend; if HOST_UID/HOST_GID in .env don't match your user (id -u / id -g), set them. The page retries every 5 seconds.
“Who's buying & selling” shows no FII/DII figures	NSE blocked the request that morning. Participant positions come from a public file and usually still appear. Press Refresh now later.
“Only N bars available … need at least 30”	Upload or backfill more history first; after that, one-day uploads work.
Sync log: “not found in bhavcopy”	The symbol is not an NSE ticker. Check the spelling (e.g. M&M, BAJAJ-AUTO, FEDERALBNK).
Sync log: “No bhavcopy for N weekday(s)”	Holidays, or today's file isn't published yet (NSE publishes around 6 pm IST). Normal.
Sync log: “Ignored … different date”	NSE served a stale file; it is retried automatically at the next sync.
Sync log: “Download failed … HTTP 403”	NSE refused the request. Try again later and check the server has internet access.
Morning briefing is empty	Click **Build daily briefing now**, or wait for 08:15 on a weekday. The backend must be running.
Briefing “sources” line shows NSE ✕	NSE blocked the request. Index data falls back to the official daily file; events may be missing until the next refresh.
“A sync is already running”	Wait for it to finish — the progress bar and log show where it is.
“cannot execute: required file not found” for pip	Old virtual environment still active — open a new terminal, recreate .venv (section 3).
Docker: permission denied writing /data	Create ~/nse-analyzer-data before the first start, and set HOST_UID/HOST_GID to your ids.
Search quality seems low; health shows embedding_backend “hashing”	The embedding model couldn't be downloaded; search still works, with lower quality. Allow internet access on first start.

13. Settings reference

Variable	Default	Purpose
NSE_DATA_DIR	~/nse-analyzer-data	Where all data is stored
AUTO_SYNC	true	Turn the daily bhavcopy sync on or off
AUTO_SYNC_TIME	18:45	Daily sync time, IST, Monday–Friday
AUTO_SYNC_ON_START	true	Catch up on missed days when the server starts
AUTO_BRIEFING	true	Build the morning briefing and weekly recap automatically
BRIEFING_TIME	08:15	Daily briefing time, IST, Monday–Friday (keep before 08:30)
WEEKLY_TIME	09:00	Weekly recap time, IST, Saturday
SENTIMENT_BACKEND	lexicon	finbert for the optional local FinBERT model

Variable	Default	Purpose
EMBEDDING_BACKEND	auto	Force sentence_transformers, onnx or hashing embeddings
CHART_BARS	750	Number of bars sent to the charts
CORS_ORIGINS	localhost:5173	Allowed UI origins when not using the proxy

API endpoints

Endpoint	Purpose
POST /api/analyze	Upload a CSV (fields: file, symbol, merge, ephemeral)
POST /api/analyze-stored/{symbol}	Analyse stored data without downloading
GET /api/results/{symbol}	Latest saved analysis
GET · POST /api/watchlist	List with latest verdicts / add symbols
DELETE /api/watchlist/{symbol}	Stop tracking a symbol
POST /api/sync · GET /api/sync/status	Run a sync / its progress and log
GET /api/history/{symbol}	All stored analyses for a symbol
GET /api/search?q=	Semantic search over stored analyses
GET /api/briefing/latest	Latest daily briefing and weekly recap
POST /api/briefing/refresh	Build now: {"kind": "daily"} or "weekly"
GET /api/health	Status, stored count, data folder

Full interactive documentation: http://localhost:8000/docs